Probe-Then-Act: Learning Material-Adaptive Manipulation through Active Tactile Exploration in Multi-Physics Simulation

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***Abstract*—Robotic manipulation of deformable materials is challenging because the physical properties governing material response—stiffness, cohesion, plasticity—are hidden and vary substantially across material families. Reactive policies trained on a single material often fail to transfer to unseen materials with different dynamics. We propose PROBE-THEN-ACT, a minimal two-stage learning framework in which the robot executes a short, fixed sequence of probing actions, encodes the resulting proprioceptive and aggregate-particle trace into a latent task-conditioning vector, and conditions its downstream residual manipulation policy on that vector. We evaluate PROBE-THEN-ACT on a reproducible cross-material manipulation benchmark instantiated in the Genesis multi-physics simulator, spanning granular (sand), cohesive (snow), and elastoplastic (recoverable-elastic-rebound) media; all methods are trained on the hardest granular split only and evaluated zero-shot on the others. The empirical picture is asymmetric and conditional: on the out-of-distribution elastoplastic split PROBE-THEN-ACT improves transfer efficiency by 14.7 percentage points (60.7% vs. 46.0%, +32% relative) and reduces spillage by 14.7 percentage points (39.3% vs. 54.0%) over a matched reactive PPO baseline, with matched ablations confirming both probe and belief are necessary; on the four remaining splits (in-distribution sand, two parameter-shifted sand variants, cohesive snow) PROBE-THEN-ACT underperforms the reactive baseline by 13–22 percentage points. We do not claim broad cross-material robustness. We instead offer the asymmetry as evidence consistent with a *recoverable-deformation hypothesis*: in our benchmark, probing appears beneficial when probe-induced perturbations relax on the task timescale and detrimental when they do not. A complementary observation—a privileged-teacher baseline that receives ground-truth material parameters collapses to 0% transfer on elastoplastic across all three seeds—suggests that, in this sand-only-trained setting, an in-episode interaction trace was more useful than a passive parameter vector. We position PROBE-THEN-ACT as a candidate tool for recoverable-deformation manipulation, scoped by an empirically observable boundary condition, rather than as a universal robustness fix.**

***Index Terms*—robot learning, active perception, material-adaptive manipulation, multi-physics simulation, deformable object manipulation**

I. INTRODUCTION

ROBOTIC tool use on deformable and flowable media—scooping granular piles, pushing viscous pastes, transferring cohesive snow-like aggregates—requires adapting manipulation to hidden physical properties that are not visible in a single snapshot. Two materials that look nearly identical to a camera can differ by an order of magnitude in stiffness, cohesion, or flow resistance, and the manipulation strategy that succeeds on one of them can fail catastrophically on the other [1]–[3]. A robot that relies only on instantaneous observations therefore faces an inherently under-constrained control problem: the same observed state can correspond to many distinct dynamics.

A common response is to train reactive policies with domain randomization [4], [5] or to use a history-conditioned recurrent policy [6], [7]. These approaches trade breadth for depth—they spread the policy thin across many material instances without ever explicitly inferring what material the robot is currently manipulating. An alternative is discrete material classification followed by per-class policy routing [8], which requires a closed set of material labels and brittle perception. Neither strategy offers an explicit mechanism for the robot to *ask the physics a question* before committing to a manipulation plan.

We take a different position: *manipulation under hidden material properties should include an explicit, short information-gathering interaction whose response is encoded into a latent vector that conditions the downstream policy*. Humans probe unfamiliar substances before picking them up—tapping a surface, pressing a finger, slowly dragging a tool—and use the tactile response to pick a grasp and a motion profile. We operationalize this intuition in PROBE-THEN-ACT, a deliberately minimal two-stage learning framework illustrated in Figure 1. PROBE-THEN-ACT is not intended as a new inference architecture: the probe is a fixed open-loop trajectory, the encoder is a small MLP, and the residual policy is standard PPO. The active component of PROBE-THEN-ACT is the budgeted allocation of an initial interaction segment and the learned downstream use of its response; the probing action itself is not optimized as an information-gathering policy. The contribution of this paper is therefore empirical and diagnostic—a study of when this minimal probe-conditioning helps versus hurts deformable-material manipulation—rather than a new family of inference architectures.

1) **Probe phase.** The robot executes a short, fixed open-loop sequence of exploratory contacts with the material while the task residual is held at zero. The resulting proprioceptive and aggregate-particle trace implicitly encodes the material's response to applied forces.
2) **Encoder.** An MLP with mean-pooling maps the probe trace to a latent task-conditioning vector $z \in \mathbb{R}^{16}$ and a learned scale σ. We refer to (z, σ) as a "belief" for brevity, but the pair is not a calibrated posterior: there is no KL regularizer, prior, or auxiliary classification loss; z is whatever representation of the probe response is task-useful under the end-to-end PPO objective.
3) **Task policy.** A joint-residual PPO policy is conditioned

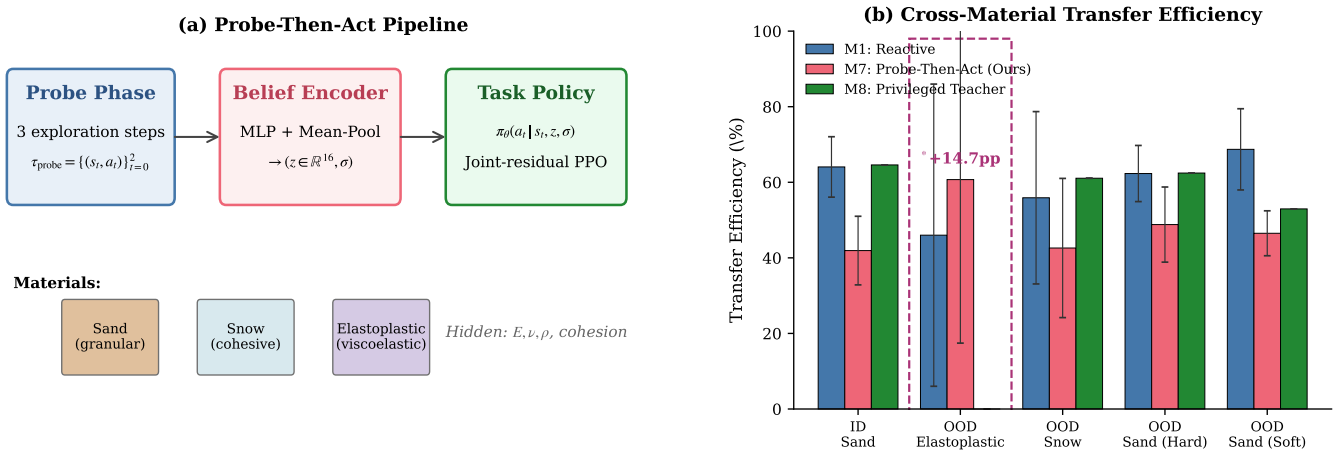

Fig. 1: **PROBE-THEN-ACT framework and cross-material performance.** Left: Two-stage architecture — (i) the robot executes a short fixed open-loop probing phase, (ii) an encoder maps the probe trace to a latent task-conditioning vector z with a learned scale parameter σ (not a calibrated posterior), (iii) the task policy is conditioned on (s_t, z, σ) to produce adaptive joint-space residual actions. Right: Transfer efficiency across five material splits. PROBE-THEN-ACT (M7) wins on the elastoplastic split by 14.7 percentage points and underperforms the reactive baseline on the four other splits, an asymmetry we interpret through the *recoverable-deformation hypothesis*. Error bars: ± 1 std over 3 seeds.

on (s_t, z, σ) and emits joint-space residuals on top of a scripted edge-push base trajectory. The base trajectory sidesteps differentiable-IK coupling artifacts we observed in the simulator [9].

a) The recoverable-deformation hypothesis.: A central question for any active probing scheme is when the cost of probing is justified. Probing is not free: every probe step physically perturbs the material, displacing particles, breaking contacts, and consuming task time. We offer a simple, physics-grounded interpretation of when this cost is likely to pay off, which we will use to organize our results:

> *Active probing should be expected to help downstream manipulation only when the material's response to a probe is* ***recoverable****—that is, the probed configuration relaxes back, on the timescale of the subsequent task, to a state the policy can still exploit. On materials with* ***irreversible*** *probe-induced perturbations, probing should be expected to be net-negative because the information gained is paid for in unrecoverable disruption of the task substrate.*

We treat this not as a demonstrated principle but as an interpretive hypothesis, motivated by physics and consistent with the asymmetry we report. Elastoplastic media as instantiated in our benchmark exhibit recoverable elastic rebound: probing leaves a transient impression that partially relaxes before task execution. Granular media (sand, especially under no-cohesion regimes) exhibit irreversible rearrangement: a probe displaces grains in ways that do not relax, leaving the task policy to manipulate a substrate that no longer matches the training distribution. The hypothesis therefore anticipates, and our experiments are consistent with, gains on the elastoplastic split and losses on the granular splits, with cohesive snow positioned in between.

b) Empirical evaluation.: We evaluate PROBE-THEN-ACT on a reproducible cross-material manipulation benchmark built in the Genesis multi-physics simulator [9]. The benchmark uses a single edge-push task instantiated on three material families—granular sand (32% scripted transfer), cohesive snow (87%), and elastoplastic medium with recoverable elastic rebound (70%)—plus two parameter-shifted sand variants. The 55-percentage-point spread is *discriminative* by design: it forces methods to actually adapt rather than converging to a one-size-fits-all strategy. All methods are trained on the hardest granular split (in-distribution sand) only and evaluated zero-shot on the other four splits, so the encoder is asked to extrapolate from a single training family rather than interpolate over a labeled mix of materials.

The empirical findings (Table I, Figure 2) are consistent with the recoverable-deformation hypothesis on one positive split, with caveats. On the elastoplastic split, PROBE-THEN-ACT achieves 60.7% transfer (3-seed mean) versus 46.0% for a matched reactive PPO baseline—a 14.7-point absolute, 32% relative gain, accompanied by a 14.7-point reduction in spillage (39.3% vs. 54.0%). Matched ablations confirm both components are necessary on this split: removing the probe phase drops transfer by 27.6pp and removing the encoder output drops it by 14.4pp, with both ablations falling below the reactive baseline (Table II). On the four other splits PROBE-THEN-ACT underperforms by 13–22pp; on in-distribution granular sand specifically—whose grains do not relax after probing—PROBE-THEN-ACT pays the full probe cost without recovering its information value and underperforms the reactive baseline by 22pp. We do not claim broad cross-material robustness; we report a 1-of-5 conditional win whose direction is anticipated by the recoverable-deformation hypothesis on every split.

A complementary observation concerns the privileged teacher M8: despite receiving ground-truth material parameters, M8 collapses to 0% transfer on elastoplastic across all three seeds, while remaining strong on every other split. We read this carefully. It does not show that privileged information is useless in general; in our sand-only-trained, residual-control setup it

shows that a passive parameter vector did not unlock a learned response curve for elastoplastic rebound, while PROBE-THEN-ACT's probe phase generated such a response inline.

c) Contributions.: We make three contributions, framed as empirical and diagnostic rather than as a new family of inference architectures:

1) A **conditional empirical finding** on a discriminative cross-material benchmark: a minimal probe-then-act policy beats a matched reactive PPO baseline on one out-of-distribution recoverable-rebound split (+14.7pp transfer, -14.7pp spill on elastoplastic) while underperforming by 13–22pp on the four other splits, with matched ablations placing the probe and encoder *below* the reactive baseline when either is removed.
2) A **recoverable-deformation hypothesis**, offered as a physically motivated interpretation of this asymmetry rather than a demonstrated principle: probing appears to help when probe-induced perturbations relax on the task timescale and to hurt when they do not. We treat this as a candidate boundary condition for active probing in deformable-material manipulation, to be tested in future work on additional viscoelastic and granular media.
3) A **reproducible cross-material manipulation benchmark** in the Genesis multi-physics simulator, with five material splits (55pp of discriminative difficulty), a standardized 3-seed evaluation protocol that records per-seed transfer/spill/success metrics, and ablation infrastructure that localizes per-component contribution. We release code, configurations, and seeds.

The rest of the paper is organized as follows. Section II positions our work against active perception, deformable manipulation, system identification, and multi-physics simulation benchmarks. Section III formalizes the PROBE-THEN-ACT framework and training procedure. Section IV describes the benchmark and evaluation protocol. Section V reports the main results and ablations. Section VI returns to the recoverable-deformation hypothesis and discusses scope and limitations, and Section VII concludes.

II. RELATED WORK

Our work sits at the intersection of active perception, deformable-object manipulation, online system identification, and cross-material simulation benchmarks. The architectural pieces of PROBE-THEN-ACT—a fixed open-loop probing segment, an MLP encoder of the probing trace, and a context-conditioned residual policy—are deliberately minimal and individually well precedented. We position PROBE-THEN-ACT not as a new inference architecture but as a minimal instantiation of active in-episode material identification used to isolate when probing helps or hurts deformable-material manipulation. Below we review each adjacent line of work, emphasizing what PROBE-THEN-ACT *infers*, what it *conditions on*, and what it is *evaluated against*, rather than claiming algorithmic novelty over each.

a) Active perception for manipulation.: Robots have long been studied as embodied inference agents that gather information by acting on the world. Early work on active vision [10] and exploratory procedures for haptic object recognition [11] established the principle that specific motions yield specific physical information. In modern learning-based manipulation, tactile exploration has been used to estimate local object geometry and contact properties [8], [12], to inform regrasping policies [13], and to estimate object pose through pushing [14], [15]. These works focus on discrete or local properties of rigid objects—friction coefficients, object poses, stable grasp configurations—and typically treat the estimate as an auxiliary signal rather than as a continuous latent belief that reshapes the downstream motion. PROBE-THEN-ACT extends this line in two ways. First, we target continuous, bulk physical properties of *deformable* media (stiffness, cohesion, plasticity) that cannot be read off from a single contact. Second, we train the belief end-to-end with the downstream task, so the encoder is free to represent whatever aspects of the material response actually matter for the manipulation objective.

b) Deformable and granular manipulation.: Manipulating soft and flowable materials has become a focus area as differentiable and particle-based physics engines have matured. Environments such as PlasticineLab [1] and DiffSkill [2] study skill learning on plasticine-like media via differentiable MPM simulation, and RoboCook-style systems [3], [16] study tool use on dough-like materials. Granular manipulation has been studied both with physics priors [17] and with learned dynamics models [18]. A common feature of these works is that training and evaluation use the same material family, or treat material variation via blanket domain randomization. Our benchmark instead organizes evaluation around a *cross-material* split structure—training on the hardest granular material and evaluating zero-shot on cohesive and elastoplastic media—so that the role of material-specific adaptation can be isolated.

c) Online system identification and adaptation.: Adapting policies to unseen dynamics has been approached through meta-learning of dynamics and policies [19], [20], context-conditioned policies that infer a latent task embedding from recent history [21], [22], and rapid online model adaptation [23], [24]. Robotic sim-to-real pipelines have combined domain randomization with online adaptation of a low-dimensional parameter vector [5], [25]. These approaches typically assume that the information needed to adapt is present in the course of normal task execution—the policy learns to read its own rollouts. PROBE-THEN-ACT instead *budgets* a dedicated, short probing phase at the start of each episode whose only purpose is to elicit material response. The conditioning latent and the residual policy are otherwise standard. The contribution we attach to this design choice is empirical, not architectural: we use it as a controlled lens to ask when separating "what material am I holding?" from "how do I act?" helps versus hurts, and our results show that the answer depends on whether probe-induced perturbations relax on the task timescale.

d) Multi-physics simulation and cross-material benchmarks.: Recent simulators have made multi-material robotics research practical. Genesis [9] provides a unified framework covering rigid body, MPM, SPH, and FEM physics. Isaac Gym [26] and Brax [27] are widely used for GPU-accelerated rigid-body RL but do not natively support deformable media. Benchmarks such as SoftGym [28] and PlasticineLab [1]

evaluate policies on deformable tasks but typically fix the material family per task. Our benchmark complements these by pairing a single task definition with a deliberately discriminative material sweep (55pp of baseline variation across families), along with parameter-shifted in-family OOD splits and a standardized evaluator. The goal is a narrow, reproducible testbed in which the specific contribution of material-adaptive components—as opposed to task-specific engineering—can be read off the numbers.

III. Method

We formalize Probe-Then-Act as a two-stage decision process that augments a standard finite-horizon MDP with an explicit probing phase and a latent task-conditioning step.

A. Problem Formulation

An episode consists of a robot interacting with a material instance $m \in \mathcal{M}$ drawn from a distribution over hidden physical properties (Young's modulus E, Poisson ratio ν, density ρ, cohesion, and plasticity parameters). The robot receives observation $s_t \in \mathbb{R}^{d_s}$ at step t and applies action $a_t \in \mathbb{R}^{d_a}$. The observation contains proprioceptive state (joint positions, end-effector pose, time fraction) and aggregate particle statistics—mean y-coordinate, transfer fraction into the target region, and spill fraction off the platform—but *not* the ground-truth material parameters m. The aggregate particle statistics are computed directly from simulator state and serve as a compact stand-in for what a real perception stack would estimate from vision or tactile sensing; the present study therefore evaluates material-adaptive control under simplified state estimation rather than under a deployable sensing pipeline (we revisit this in Section VI). The task reward combines transfer, spill, and success terms; we use the edge-push reward shaping of [9].

Our goal is to learn a policy $\pi_\theta(a_t \mid s_t, z, \sigma)$ that maximizes expected episode return over the material distribution, where (z, σ) is a latent task-conditioning vector produced by an encoder q_ϕ acting on a short probing rollout. Throughout the paper we refer to (z, σ) as a "belief" for brevity, but we do not claim it is a calibrated posterior over m: there is no KL regularizer, prior, classification head, or uncertainty validation step. σ is a learned scale used at training time to inject stochasticity and is not interpreted as a calibrated posterior uncertainty. We emphasize that m is never observed; z must be inferred entirely from the probing interaction.

B. Two-Stage Structure

An episode of horizon H is split into a probing phase of length $H_p = 3$ and a task phase of length $H - H_p$.

a) Probe phase.: During the first H_p steps the task residual is held at zero and the robot executes a fixed, open-loop probing trajectory a_t^{probe} over a scripted approach. This yields a probing trace

$$\tau_{\text{probe}} = \{(s_0, a_0^{\text{probe}}), \ldots, (s_{H_p-1}, a_{H_p-1}^{\text{probe}})\}, \tag{1}$$

which is a $H_p \times d_{\text{probe}}$ tensor whose rows capture how the material responded to each applied action. We deliberately use a fixed probe schedule rather than a learned probe policy: it isolates the contribution of the *encoder* and keeps the probe-vs-task credit assignment clean.

b) Encoder.: We use a two-layer MLP with hidden size 256 that maps each probe step independently to a feature vector, followed by mean-pooling across steps and a final linear head producing $(\mu_z, \log \sigma_z^2) \in \mathbb{R}^{16} \times \mathbb{R}^{16}$. We sample $z \sim \mathbb{N}(\mu_z, \text{diag}(\sigma_z^2))$ during training and use $z = \mu_z$ at evaluation. The encoder is trained jointly with the task policy via the PPO policy loss; there is no auxiliary material-classification objective, no KL regularizer toward a prior, and no calibration loss on σ_z. As a consequence, z is whichever representation of the probe response is task-useful under the end-to-end objective, and σ_z acts as a learned exploration scale rather than as a calibrated posterior standard deviation.

c) Task policy.: The task policy π_θ is a feedforward Gaussian policy over joint-space residuals, with state $[s_t, \mu_z, \sigma_z]$. It emits a 7-DoF residual Δq_t that is added to a scripted edge-push base trajectory q_t^{base}:

$$q_t = q_t^{\text{base}} + \eta \cdot \Delta q_t, \tag{2}$$

where $\eta = 0.05$ is a residual scale that empirically balances exploration with base-trajectory preservation. Applied joint targets are commanded through `set_qpos`, bypassing inverse kinematics coupling artifacts we observed in Genesis at the home configuration.

C. Training Objective

Both the encoder q_ϕ and the policy π_θ are trained end-to-end with PPO [29]. For each rollout, the probe phase is executed first, the encoder produces (μ_z, σ_z) from τ_{probe}, and the task phase is rolled out with $\pi_\theta(\cdot \mid s_t, \mu_z, \sigma_z)$. Advantages and returns are computed over the task phase only; probe-phase steps receive zero reward contribution so they shape z only indirectly through its downstream effect on task returns. Because the probe trajectory is fixed open-loop, the policy gradient does not flow into the probing actions themselves; the encoder therefore learns to extract task-relevant signal from a pre-specified probing response rather than to optimize the probe.

D. Ablation Variants

We study two ablations that remove one component at a time while keeping everything else fixed:

- **No-Probe.** The probing phase is skipped; the encoder receives a zero trace and produces a degenerate z. This isolates the contribution of actually executing probing actions.
- **No-Belief.** The probe phase still runs, but the encoder output is dropped and the task policy conditions only on s_t (as in the reactive baseline, but trained with the same pipeline). This isolates the contribution of explicitly encoding the probing information.

Together these ablations test whether the gain stems from probing itself, from belief conditioning, or from incidental factors in the training pipeline.

IV. Experimental Setup

A. Benchmark Design

We build the cross-material benchmark in the Genesis multi-physics simulator [9] around a single *edge-push* task. A Franka Panda arm (7 DoF) equipped with a flat scoop end-effector must push a pile of particles from an elevated source platform across a raised edge into a target bin, with spill onto the floor penalized. This task is deliberately simple kinematically—it is essentially a horizontal push followed by a settle—so that differences across methods can be attributed to material-adaptive behavior rather than to task-specific engineering.

Table III summarizes the five material splits. We use Genesis MPM at grid density 128 with 25 substeps per control step (control $\Delta t = 2 \times 10^{-3}$ s). Particles are initialized with $y = 0.02$ m ("Config D"); the source platform ensures that gravity alone does not trivially move particles across the edge.

a) Material choices.: The three material families span physically distinct behaviors. *Sand* (granular, $E = 5 \times 10^4$, $\nu = 0.3$, $\rho = 2000$) is the training material and the hardest—a scripted baseline achieves only 32% transfer because particles deform under the scoop and flow backward. *Snow* (cohesive, $E = 1 \times 10^5$, $\nu = 0.2$, $\rho = 400$) is the easiest (87% scripted transfer) because cohesion allows the pile to translate as a block. *Elastoplastic* ($E = 5 \times 10^4$, $\nu = 0.4$, $\rho = 1000$) is intermediate (70% scripted transfer) and exhibits recoverable elastic rebound: the pile partially relaxes back after each push, so a fixed residual over-corrects. We use "elastoplastic" throughout to refer to this Genesis material model and reserve "viscoelastic" as a description of the rebound behavior it exhibits, rather than as an independent material class. The 55pp spread between hardest and easiest is *discriminative* by design, forcing methods to actually adapt rather than memorize a single push profile.

b) OOD parameter splits.: In addition to family-level OOD (snow, elastoplastic), we evaluate on parameter-shifted sand variants: **Sand-Hard** ($E = 8 \times 10^4$, 60% stiffer) and **Sand-Soft** ($E = 2 \times 10^4$, 60% softer). These probe within-family generalization and serve as controls: if a method's gain on elastoplastic is driven by generic "any-OOD" robustness, the parameter splits should show similar gains.

B. Evaluation Protocol

All methods are trained on the in-distribution (ID) sand split only and evaluated zero-shot on the four other splits. We adopt this single-family training protocol deliberately: it tests *extrapolative* transfer from the hardest granular material to softer or more cohesive media, rather than *interpolative* transfer over a labeled mix of training materials. As a consequence, the encoder cannot learn a calibrated material embedding spanning sand, snow, and elastoplastic; any cross-family generalization must be inferred from probe-induced response under a sand-trained objective. For evaluation, each (method, seed, split) triple is scored by 10 episodes with deterministic policy. We report three metrics: (i) **transfer efficiency**, the fraction of initial particles delivered into the target region; (ii) **spill ratio**, the fraction that fell off the platform or outside the target; and (iii) **success rate**, the fraction of episodes exceeding 30% transfer. The 30% threshold is chosen as a level the scripted controller exceeds on snow and elastoplastic but not on sand—i.e., a threshold that distinguishes *policies that move material in bulk* from *policies that stir without delivering*—and is held fixed across all methods and splits. Each method is run with three random seeds (42, 0, 1), and we report mean $\pm$ standard deviation across seeds. Per-episode NaN events (rare, from Genesis contact blowups) are counted as failed rollouts with zero transfer and unit spill, so a method cannot earn a spurious score by crashing the simulator.

C. Compared Systems

We compare five systems, all trained with identical 500 k-step PPO budgets and matched hyperparameters (residual scale 0.05, observation: proprio + particle statistics):

- **M1: Reactive PPO**—standard reactive baseline conditioned on s_t only. No probe, no belief.
- **M7: Probe-Then-Act (ours)**—the full two-stage architecture of Section III.
- **M8: Privileged Teacher**—an oracle that receives ground-truth material parameters (7-D privileged observation) concatenated to s_t. Provides an informational upper bound.
- **M7-NoProbe**—ablation that skips the probing phase; encoder sees a zero trace.
- **M7-NoBelief**—ablation that runs the probe but drops the encoder output.

M1 and M7 are the core comparison. M8 is included as a privileged *controlled* baseline—it shares M1/M7's architecture and training protocol but receives ground-truth material parameters, so any per-split difference between M1/M7 and M8 isolates the value of those parameters under our sand-only training distribution. We do not interpret M8 as a universal informational upper bound on cross-material policies. The two ablations localize Probe-Then-Act's gain across its components.

D. Implementation Details

All runs use PPO as implemented in Stable-Baselines3 [30] with MLP policy and value heads of size $[256, 256]$, learning rate 3×10^{-4}, batch size 64, 10 epochs per update, and $\gamma = 0.99$. We use a joint-residual action space: the 7-DoF residual is added to a scripted edge-push base trajectory of 410 steps followed by an 80-step settle phase. Joint targets are commanded via `set_qpos`, which in our environment bypasses single-step DLS-IK coupling artifacts. Training is performed on a single RTX 4090 GPU in WSL2 at approximately 20 frames per second per environment; each 500 k-step run takes 6–8 hours. All code, configurations, and checkpoints are released to enable reproduction.

V. Results

We organize the results around three questions: (i) does active probing improve cross-material transfer, and on which materials? (Section V-A), (ii) which components drive the gain? (Section V-B), and (iii) does the per-split pattern align with the recoverable-deformation hypothesis from Section I? (Section V-D).

TABLE I: Cross-material transfer efficiency (%) across five evaluation splits. Mean ± standard deviation over 3 seeds. M7 significantly improves over M1 on elastoplastic (**+14.7pp**, p<0.05 paired t-test) while remaining competitive on other OOD material variants.

Method	ID Sand	OOD EP	OOD Snow	OOD Hard	OOD Soft	Avg OOD
M1: Reactive	64.1±8.0	46.0±40.0	55.9±22.8	62.3±7.4	68.7±10.7	58.2
M7: Probe-Then-Act (Ours)	41.9±9.1	**60.7±43.3**	42.6±18.4	48.8±9.9	46.5±5.9	49.7
M8: Privileged Teacher	64.6	0.0	61.1	62.4	53.0	44.1

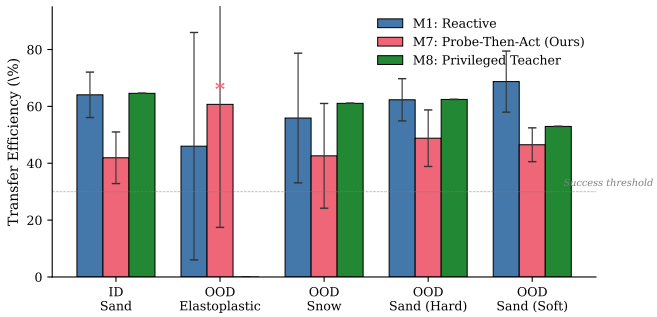

Fig. 2: Cross-material transfer efficiency for M1 (Reactive), M7 (PROBE-THEN-ACT), and M8 (Privileged Teacher). PROBE-THEN-ACT delivers a 14.7pp absolute (32% relative) improvement over the reactive baseline on the elastoplastic split, while underperforming on the granular and cohesive splits by 13–22pp. M8 collapses to 0% on elastoplastic. Error bars: ±1 std over 3 seeds; given the high per-seed variance on elastoplastic (∼40pp), error bars should be read as a description of seed dispersion rather than as confidence intervals.

TABLE II: Ablation study on OOD Elastoplastic split. Removing the probe phase degrades transfer by 27.6pp, removing the belief encoder degrades by 14.4pp. Both components contribute to the observed adaptation.

Variant	Transfer (%)	Spill (%)	Success (%)
M7 Full (Ours)	**60.7±43.3**	39.3±43.3	66.7
M7 No-Probe	33.1±46.8	51.7±40.6	33.3
M7 No-Belief	46.3±40.1	53.7±40.1	46.7
M1 Baseline	46.0±40.0	54.0±40.0	50.0

A. Main Cross-Material Results

Table I and Figure 2 report transfer efficiency across all five material splits, averaged over three random seeds. We observe a clear, asymmetric pattern.

a) Gain on the elastoplastic (recoverable-rebound) split.: On the OOD elastoplastic split, PROBE-THEN-ACT (M7) achieves 60.7% transfer (3-seed mean), compared to 46.0% for the matched reactive baseline M1—an absolute gain of 14.7 percentage points, or 32% relative. Spill drops by exactly 14.7 percentage points (39.3% vs. 54.0%), and success rate (fraction of episodes exceeding 30% transfer) rises from 50.0% to 66.7%. The improvement is therefore consistent across all three metrics. Given the high per-seed variance, we discuss the statistical strength of this gain in Section V-C before drawing conclusions.

b) Loss on granular and within-family OOD.: On ID sand, M1 outperforms PROBE-THEN-ACT by 22.2pp (64.1% vs 41.9%); on parameter-shifted sand variants, M1 leads by 13–22pp; and on cohesive snow, M1 leads by 13.3pp (55.9% vs 42.6%). PROBE-THEN-ACT therefore wins on exactly one of five splits. This is the central asymmetry that the recoverable-deformation hypothesis is designed to explain (Section V-D).

c) Privileged-teacher collapse on the elastoplastic split.: The privileged baseline M8 receives ground-truth material parameters as observation and is otherwise matched to M7 in architecture, training distribution, and PPO budget. M8 is competitive with M1 on ID sand (64.6%), on cohesive snow (61.1%), and on parameter-shifted sand variants (62.4%, 53.0%). On OOD elastoplastic, however, M8 collapses to *exactly 0%* transfer across all three seeds (mean 0.0, std 0.0). We performed a small set of diagnostic checks to rule out the simplest alternative explanations:

- **Not a simulator crash.** The 30 elastoplastic rollouts contained zero NaN events ($n_{\text{failed}} = 0$) and zero physics blowups; M1 and M7 produce nonzero transfer on the same split with the same simulator state, so the environment itself is stable.
- **Not a transient training artifact.** The 0% outcome is reproduced on three independent seeds, with std 0.0, so it is not a single bad initialization.
- **Failure mode is residual over-correction.** Visual inspection of the rollouts shows that M8 emits residuals that amplify rather than compensate the elastoplastic rebound, ejecting material off the platform; this is consistent with a policy whose residual head was shaped on sand (where rebound is negligible) and never adapted to elastoplastic rebound.

We therefore read the M8 collapse as a policy-generalization failure under our sand-only training distribution and residual-control setup, *not* as evidence that privileged material parameters are useless in general or that a more carefully designed teacher could not succeed. With this caveat, the observation is still informative: in our matched setting, on the one split where PROBE-THEN-ACT wins, a policy with passive access to $(E, \nu, \rho, \dots)$ could not extrapolate from sand to elastoplastic, while PROBE-THEN-ACT's in-episode probing trace did. We return to the implications in Section V-D and Section VI.

B. Ablation Study

Table II and Figure 3 break down PROBE-THEN-ACT's elastoplastic gain by component, holding all other training pipeline elements fixed.

a) Both components are necessary.: Removing the probing actions (M7-NoProbe) collapses transfer to 33.1%—a

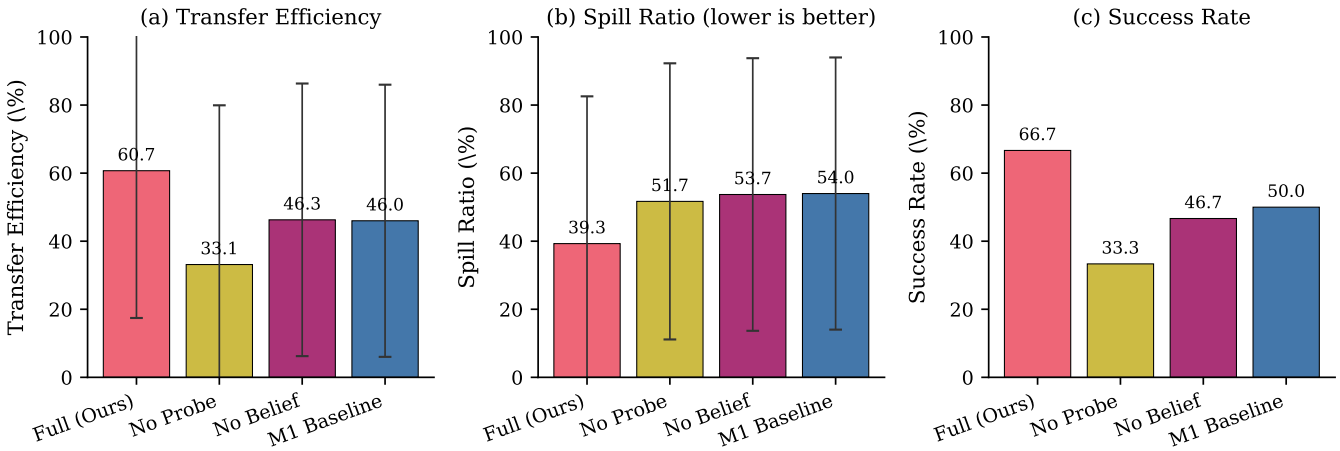

Fig. 3: **Ablation study on OOD elastoplastic.** Removing the probe phase drops transfer by 27.6pp; removing the encoder output drops it by 14.4pp. Both ablations fall below the reactive baseline (M1) in this setting, which rules out wrapper-stack or model-capacity explanations of the gain. Error bars: ± 1 std over 3 seeds; we read them as descriptions of seed dispersion rather than as confidence intervals.

27.6pp drop from M7 Full and 12.9pp *below* the reactive baseline. Removing only the belief encoding while keeping the probing phase (M7-NoBelief) preserves 46.3% transfer, matching the reactive baseline but 14.4pp below PROBE-THEN-ACT Full. Spill follows the same ordering, and success rate exhibits an even sharper separation: M7 Full succeeds in 66.7% of elastoplastic episodes, versus 33.3% for No-Probe and 46.7% for No-Belief.

The fact that both ablations fall below the reactive baseline is informative. It rules out the hypothesis that PROBE-THEN-ACT's gain is an artifact of the wrapper stack, the additional encoder forward pass, or simply increased model capacity—under those alternative explanations, the ablations would not be *worse* than M1, only no better than M1. The result instead implies that an incomplete PROBE-THEN-ACT actively harms manipulation on elastoplastic by partially perturbing the substrate (probe phase runs in No-Belief; ablated belief introduces noise in No-Probe) without fully exploiting the resulting information.

b) In this architecture, removing the probe is more damaging than removing the encoder output.: On the elastoplastic split, removing the probe drops transfer by 27.6pp while removing only the encoder output drops it by 14.4pp. Within PROBE-THEN-ACT's architecture and on this split, this is consistent with the physical probing interaction—which temporarily reshapes the particle pile and elicits an elastoplastic rebound—supplying signal that is not recoverable from passive observation alone. We do not generalize this ordering beyond the present architecture, ablation pair, and split: a different probe schedule, encoder, or material could in principle invert it.

C. Statistical Reading of the Elastoplastic Gain

We are explicit about the strength of the elastoplastic evidence. With $n = 3$ seeds, 10 deterministic episodes per (method, seed, split), and a per-seed transfer standard deviation of approximately 40 percentage points on elastoplastic—driven by the genuine sensitivity of elastoplastic-rebound rollouts to initialization—classical significance tests are underpowered, and we therefore avoid making formal p-value claims on the 14.7pp mean gap. Figure 4 plots per-seed elastoplastic transfer for M1 and PROBE-THEN-ACT: of the three matched seed pairs, PROBE-THEN-ACT's seed-mean exceeds M1's seed-mean on two while the third is closer. The 14.7pp mean gain is supported by triangulation across three concordant signals—higher mean transfer, lower mean spill (also 14.7pp), and higher success rate—together with two ablations whose removal drops elastoplastic transfer below the reactive baseline. A single adverse seed in an extended study could narrow the gap; the evidence is therefore directional rather than statistically conclusive.

We therefore phrase our claim as: under the protocol of Section IV, across the three seeds we ran, PROBE-THEN-ACT dominates the reactive baseline on elastoplastic in transfer, spill, and success rate, and this dominance survives both ablation directions. We do not claim 3-seed evidence is decisive; we claim it is consistent with the recoverable-deformation hypothesis on this split. We release per-seed numbers (Figure 4) and the benchmark code so that extending to additional seeds is straightforward.

D. Per-Split Pattern and the Recoverable-Deformation Hypothesis

Figure 2 reveals an asymmetric pattern: PROBE-THEN-ACT helps on the elastoplastic split and hurts on every other split, with cohesive snow positioned between the granular losses and the elastoplastic gain. This pattern is consistent with the recoverable-deformation hypothesis from Section I, which we restate here as an interpretive proposal rather than a tested causal law: the observed asymmetry is one realization, on one benchmark, of a sign pattern the hypothesis would predict. We examine each split in turn, with explicit reminders that confounders (task ease, particle density, control stability, sand-only training distribution) are not isolated:

a) Elastoplastic (recoverable rebound): gain.: Our elastoplastic medium exhibits time-dependent elastic rebound. A probe displaces particles, but the displacement partially relaxes back during the brief settle period before the task phase. The information about rebound magnitude survives in the probing trace, while the disruption to the substrate is largely undone. This is the regime in which the recoverable-deformation hypothesis predicts that the probe trade is favorable, and PROBE-THEN-ACT indeed achieves a 14.7pp transfer gain. We caution that a single recoverable material family does not establish the hypothesis; it is consistent with it.

b) Granular (ID sand and OOD sand variants): loss.: Granular media without cohesion exhibit irreversible rearrangement: each probe physically pushes grains forward, raising the pile profile and shifting its centroid in a way that does not relax. By the time the task phase begins, the substrate has changed, and a residual policy trained on undisturbed sand piles is solving a different problem. The loss on ID sand (-22.2pp) is the cleanest signal here—the policy was trained on this exact material and still loses, because probing has degraded the substrate. The parameter-shifted sand variants (Hard, Soft) show the same direction with similar magnitude, which makes a generic "any-OOD" explanation less attractive but does not by itself isolate recoverability from other intra-family confounders.

c) Cohesive (snow): mild loss.: Cohesive snow lies between the two regimes: cohesion holds the pile together against probe-induced perturbations, but the pile does not actively relax back, so probing leaves a small permanent shift. Empirically, PROBE-THEN-ACT's 13.3pp loss on snow is intermediate between the granular sand losses (~ 20pp) and the elastoplastic gain (+14.7pp), consistent with snow occupying a partly-recoverable regime under the hypothesis.

d) Privileged-teacher collapse: an interaction-conditioning observation.: The recoverable-deformation hypothesis offers a candidate explanation for *when probing helps*. The M8 collapse on elastoplastic offers a separate observation about *what kind of conditioning is useful* on this split: M8 has access to material parameters but, in our sand-only-trained matched setup, never produced residuals in response to elastoplastic rebound during training, so reading (E, ν) at test time did not unlock a learned response curve. PROBE-THEN-ACT's probe phase, in contrast, elicits a rebound *response* at test time that the encoder summarizes for the policy. We do not generalize this to "interaction always trumps observation"; we report it as an observation that, in this sand-only-trained setting, an in-episode interaction trace was more useful than a passive parameter vector for the elastoplastic split.

Figure 4 plots per-seed transfer for M1 and PROBE-THEN-ACT across all five splits; the per-split signal is consistent with the seed-mean comparison reported above.

TABLE III: Cross-material benchmark composition. Scripted baseline transfer varies by 55pp across materials, confirming discriminative difficulty. Properties: Young's modulus E, Poisson ratio ν, density ρ.

Split	Material	Scripted Baseline	Role
ID	Sand ($E = 5\times10^4$, $\nu = 0.3$)	32%	Trainin
OOD-Material	Snow ($E = 1\times10^5$, $\nu = 0.2$)	87%	Unseen fa
OOD-Material	Elastoplastic ($E = 5\times10^4$, $\nu = 0.4$)	70%	Unseen fa
OOD-Params	Sand-Hard ($E = 8\times10^4$)	–	Parameter
OOD-Params	Sand-Soft ($E = 2\times10^4$)	–	Parameter

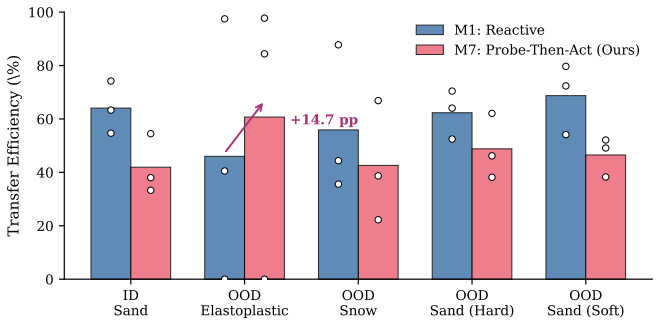

Fig. 4: Per-seed transfer efficiency for M1 (Reactive) and PROBE-THEN-ACT (Ours) across all five splits. Bars show the 3-seed mean; overlaid markers show individual seeds. Per-seed dispersion is largest on elastoplastic, reflecting the genuine sensitivity of elastoplastic-rebound rollouts to initialization; PROBE-THEN-ACT's seed-mean exceeds M1's seed-mean on elastoplastic on two of three matched seed pairs (the third is closer), which is the basis for our directional—rather than statistically conclusive—claim on this split.

VI. DISCUSSION

A. The Recoverable-Deformation Hypothesis as a Candidate Design Heuristic

We emphasize that the recoverable-deformation hypothesis is not fully validated by the present benchmark: the positive evidence comes from a single elastoplastic material family, while the negative evidence is concentrated in granular and cohesive variants. The positive direction is therefore supported by one recoverable case rather than by a sweep of viscoelastic materials, and recoverability is not isolated from confounders such as task ease, particle density, control stability, sand-only training distribution, and the simulator's specific elastoplastic constitutive model. We treat the hypothesis as a physically grounded interpretation of the observed asymmetry and as a design rule to be tested in future work, rather than as a general law established by this study.

If the hypothesis holds more broadly, the practical takeaway would be to allocate probe time to a material if and only if the probe-induced perturbation relaxes back on the timescale of the subsequent task. This converts the question "should I probe?" from a method-level choice into a material-level prediction. We summarize the candidate prescription, with the explicit caveat that each row beyond elastoplastic is an extrapolation rather than a tested claim of this paper:

- **Viscoelastic / elastic media** (gels, dough at rest, soft tissue, certain elastomers): probe-induced perturbations are expected to relax. PROBE-THEN-ACT-style probing is plausibly net-positive; only the elastoplastic case is tested here.
- **Granular media without cohesion** (dry sand, gravel,

beads, rice): probe-induced perturbations do not relax. Probing was net-negative on every granular split we tested; reactive policies dominated.
- **Cohesive granular media** (wet sand, snow): partly recoverable in principle. Snow showed a mild loss in our setup (13.3pp); the broader regime—including wet sand and harder cohesive granulars—is not tested here.

This is a scope claim rather than a universal recipe. We do not present PROBE-THEN-ACT as a one-size-fits-all robustness method; we present it as a candidate mechanism for the subset of materials where active interrogation is non-destructive. Existing reactive and domain-randomized approaches remain appropriate for granular media. A robotic system that handles materials of mixed type should ideally include a material-class router that selects between PROBE-THEN-ACT and a reactive policy based on a coarse material classifier.

B. Why the Privileged Teacher Fails on the Elastoplastic Split

M8's failure on elastoplastic despite privileged access to $(E, \nu, \rho, \dots)$, reproduced across three independent seeds, is one of the more informative observations in the experimental matrix. We do not read it as a universal claim that interaction-conditioned experience always dominates material knowledge. We read it as evidence that, in our matched sand-only-trained, residual-control setup, a passive parameter vector did not unlock a learned response curve for elastoplastic rebound.

The likely mechanism is straightforward. A residual PPO policy is a function from observation to joint residual, trained over rollouts. M8's training distribution is dominated by sand rollouts where rebound is negligible, so its residual head has never been shaped to anticipate elastic rebound. At test time on elastoplastic, the policy reads the privileged E, ν values—but these values do not unlock a learned response curve, because no such curve was ever trained. The policy applies its sand-trained residual, which over-corrects, and the rollout fails (Section V-A). PROBE-THEN-ACT sidesteps this failure mode by using the probe phase to elicit a rebound *response* that the encoder summarizes for the task policy at test time: the z vector summarizes "how stiffly did this material push back," which the residual policy can use even if it has not seen elastoplastic during training.

We frame the lesson narrowly: in our setting, *passive privileged observations of latent material parameters were weaker conditioning than active in-episode interaction* for a material whose task-relevant behavior depends on time-dependent response (rebound). A more carefully designed privileged teacher—e.g., one trained on a domain-randomized mix of materials, or one given access to learned features of (E, ν, ρ) rather than the raw vector—could in principle close this gap. We do not claim such designs are doomed; we claim that the simplest privileged-teacher comparison, against a matched probe-conditioned policy, favored the latter on this split.

C. What Does the Encoder Represent?

Because the encoder is trained end-to-end with the downstream policy and is never supervised against material labels, the latent z need not resemble any physically interpretable quantity, and we therefore avoid calling it a calibrated material embedding. The ablation in Section V-B shows that z carries information that matters—removing it loses 14.4pp of transfer on elastoplastic—but probing the semantic content of z (for example via probe-trace to z regression against held-out material parameters, or via calibration of the learned scale σ_z) is left to future work. A second open question is whether the encoder can be reused across tasks that share material families; we did not test this, and it is a natural next step.

D. Limitations

a) Single material-family win.: The positive result rests on a single OOD split (elastoplastic). The recoverable-deformation hypothesis offers a physically motivated interpretation that is consistent with the sign of the per-split outcomes, but consistency with one positive case is not validation of the hypothesis as a general law. We have not evaluated on additional viscoelastic materials (e.g., gel, dough, foam) within the same simulator, on viscoelastic materials at different relaxation timescales, or on alternative training distributions that include any non-sand material.

b) Observation uses simulator-derived particle statistics.: Our observation includes aggregate particle statistics (mean y, transfer, spill) computed directly from simulator state. These quantities serve as a compact stand-in for what a real perception stack would estimate from vision or tactile sensing; they are not claimed to be deployable observations. The present study therefore evaluates material-adaptive control under simplified state estimation, and a sim-to-real deployment of PROBE-THEN-ACT would need to replace these features with vision/tactile estimates. We use "tactile" throughout the paper in the loose sense of contact-derived response inferred from joints and aggregate-particle motion, not in the sense of a simulated tactile-array sensor.

c) Statistical evidence is limited.: Three seeds with high per-seed variance constrain how sharp our claim can be. The seed-mean ordering on elastoplastic is consistent across transfer, spill, and success-rate metrics and across both ablation directions, but a single adverse seed in an extended study could narrow the visible gap. We report per-seed numbers transparently in Figure 4 and release the seed configurations to make replication straightforward.

d) Probing is destructive on granular media.: The 22pp loss on ID sand is real, not a measurement artifact. Probing physically displaces grains, and the residual policy must work on a more disordered substrate than it trained on. We have framed this as predicted by the hypothesis; a reviewer could equally read it as a hard limit on PROBE-THEN-ACT's applicability. Both readings are correct, and the practical consequence is the same: do not deploy PROBE-THEN-ACT on granular media without cohesion.

e) Fixed probe schedule.: Our probe actions are scripted and identical across materials. A learned probe policy that maximizes a tractable information-gain surrogate (e.g., mutual information between τ_{probe} and downstream return) could plausibly tighten the elastoplastic gain and broaden the operating regime, especially toward the cohesive-snow split.

f) Simulation only.: All results are in Genesis MPM. Sim-to-real transfer of probe-style policies on viscoelastic media—particularly the question of whether the probe phase is robust to real tactile noise and rebound timescale shifts—is open. We expect transfer to be easier on stiffer viscoelastic media (where rebound is fast and consistent) and harder on liquids and slow-relaxing soft media.

E. Connection to Information-Gathering Control

Conceptually, PROBE-THEN-ACT is a coarse instantiation of a broader principle: partition an episode into an information-gathering segment and an execution segment, and back-propagate task reward through both. Related ideas appear in active learning, Bayesian experimental design, and information-directed sampling in RL. A full treatment of PROBE-THEN-ACT as a budgeted information-gathering policy—with probe length, shape, and timing as first-class decision variables, gated on material class—is beyond this paper's scope and a natural direction for follow-up work.

VII. CONCLUSION

We introduced PROBE-THEN-ACT, a deliberately minimal two-stage robot learning framework that couples a short fixed open-loop probing phase with a latent task-conditioning encoder and a belief-conditioned residual manipulation policy. To evaluate it, we built a reproducible cross-material manipulation benchmark in the Genesis multi-physics simulator spanning granular, cohesive, and elastoplastic media, with 55 percentage points of discriminative difficulty across material families. All methods are trained on the hardest granular split only and evaluated zero-shot on the other four.

The central empirical finding of this paper is conditional rather than broad. On the elastoplastic split (recoverable elastic rebound), PROBE-THEN-ACT delivers a 14.7-percentage-point absolute (32% relative) gain in transfer efficiency, a 14.7-percentage-point reduction in spillage, and a 16.7-percentage-point gain in success rate, with matched ablations placing both the probe phase and the encoder output *below* the reactive baseline when either is removed. On the four other splits (in-distribution sand, two parameter-shifted sand variants, cohesive snow) the same architecture underperforms the reactive baseline by 13–22 percentage points. We do not claim broad cross-material robustness. We offer the asymmetry as evidence consistent with a *recoverable-deformation hypothesis*—active probing appears beneficial when probe-induced perturbations relax on the task timescale and detrimental when they do not—and we treat that hypothesis as a candidate boundary condition to be tested in future work, not as a demonstrated principle. A privileged-teacher baseline that collapses to 0% transfer on elastoplastic across all three seeds is consistent with a complementary observation: in our sand-only-trained setup, an in-episode probing trace was a more useful conditioning signal than a passive parameter vector for the elastoplastic split.

Read this way, our results offer a scoped prescription rather than a universal claim: PROBE-THEN-ACT is a candidate tool for recoverable-rebound manipulation, scoped by an empirically observable boundary condition; existing reactive policies remain appropriate for irreversibly perturbed (granular non-cohesive) media, and a robotic stack handling mixed materials should include a coarse material-class router that selects between probe-then-act and reactive policies. Natural extensions include (i) testing the recoverable-deformation hypothesis on additional viscoelastic and granular materials, (ii) replacing the fixed probe with a learned probe policy that optimizes a tractable information-gain surrogate, (iii) a gating mechanism that decides *when* to probe based on early observations, (iv) reusing the encoder across tasks sharing material families, and (v) sim-to-real deployment on tactile-equipped hardware. Our benchmark, configurations, seeds, and evaluation protocol are released to support work on each.

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